

OpenClaw Scenario Sandbox Comprehensive Case

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1. Course Content

- This section uses OpenClaw to control the robot in the sand table map scenario to complete complex long-flow tasks.

2. Preparation

2.1 Complete Prerequisite Courses

- Complete the [02-Build Sand Table Map], [03-Visual Relocation Markers], and [04-Map Mapping Markers] courses.

2.2 Start the MicroROS Chassis Agent

- Skip if already started

```
sh start_agent.sh
```

2.3 Start the Road Network Container

```
roadnet start
```

2.4 Start Odometry, tf, Robotic Arm Assistant, Camera, ArUco Detection Nodes

```
ros2 launch m3pro_bringup car_base.launch.py relocation:=True
```

2.5 Start the MCP Service

```
ros2 launch m3pro_bringup mcp_service.launch.py
```

2.6 Start Road Network Navigation

- Start inside the roadnet container

```
roadnet bash
```

```
ros2 launch road_net_route roadnet_nav2.launch.py  
rviz_config_file:=/root/volumes/code_ws/src/road_net_route/rviz/nav2_relocaton.rviz
```

2.7 Voice Function Nodes

- If you need to use voice interaction, you need to additionally start the following nodes, otherwise you can ignore them.
- Voice recognition node

```
ros2 run multi_brains_pre asr_detect
```

- Voice reply node

```
ros2 run multi_brains_pre openclaw_bridge
```

3. Reference Cases

[!TIP]

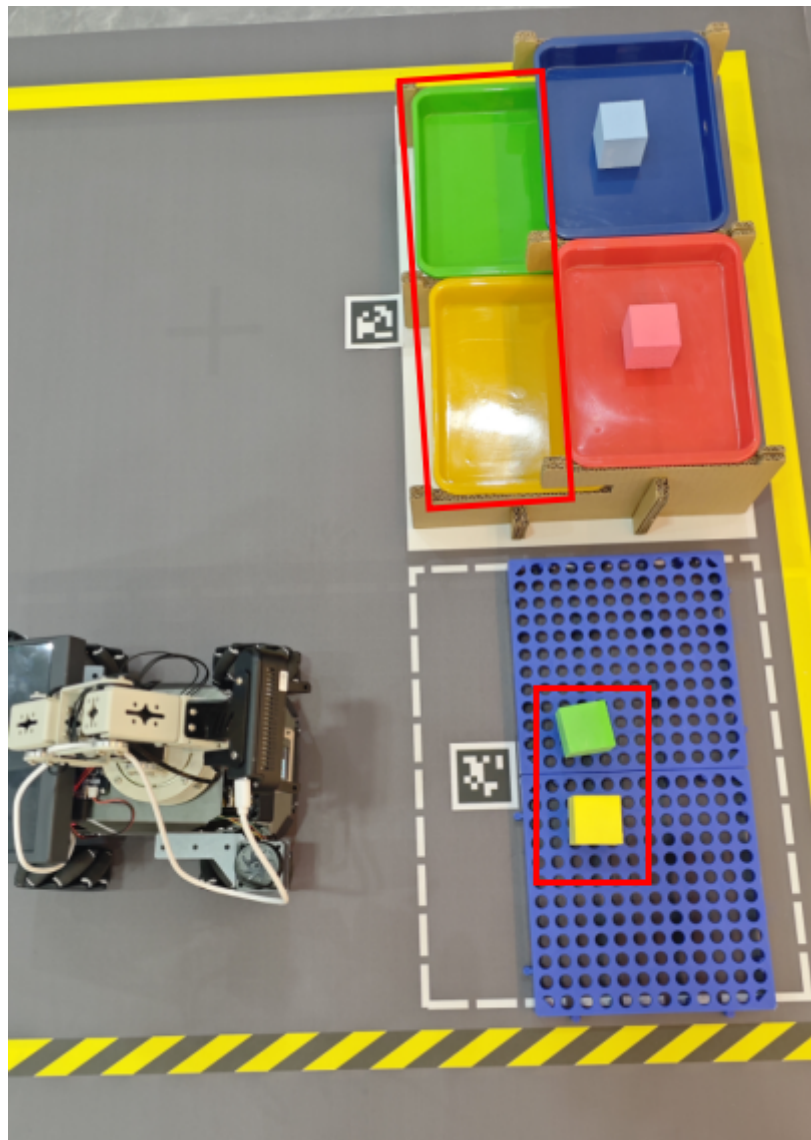
- You can customize your own command cases and gameplay features according to actual needs. The following are some reference cases.

3.1 Garbage Sorting

- Sort specified garbage from the garbage sorting area into corresponding trash bins.

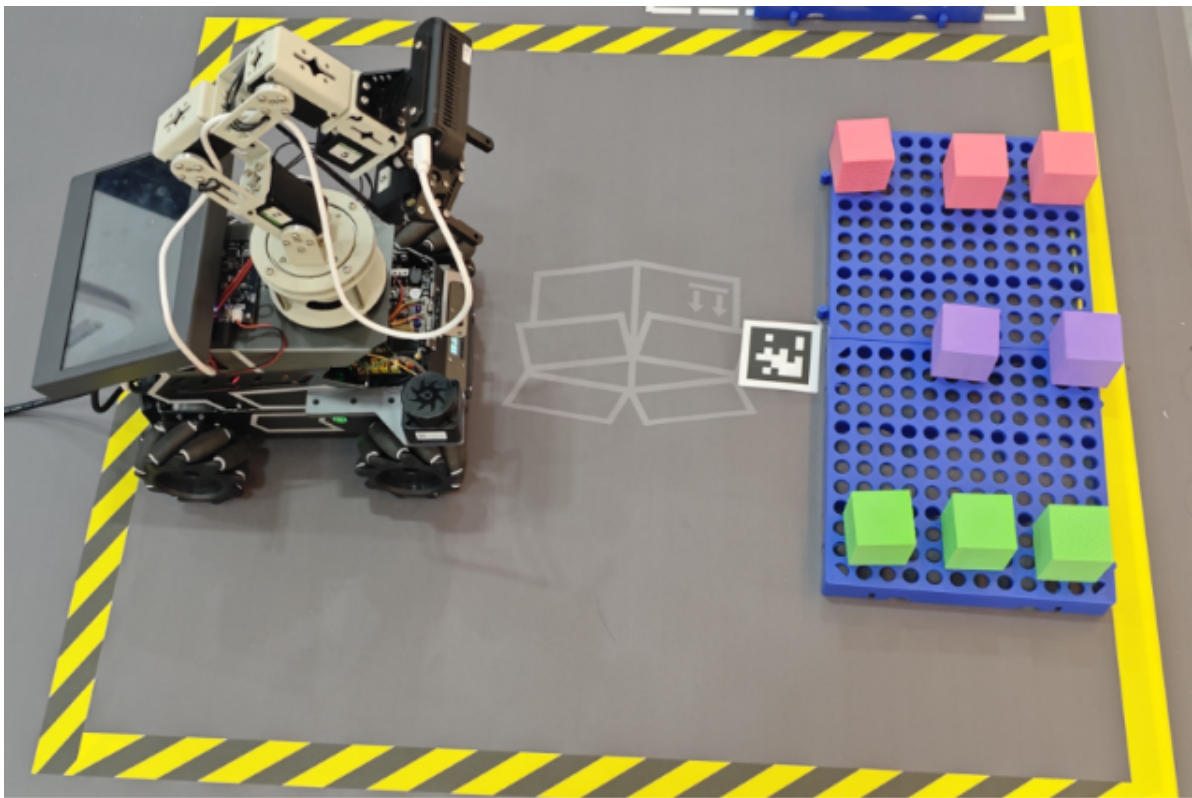
3.2 Cargo Sorting

- Sort all colors of cargo from the cargo sorting area to the designated shelves.

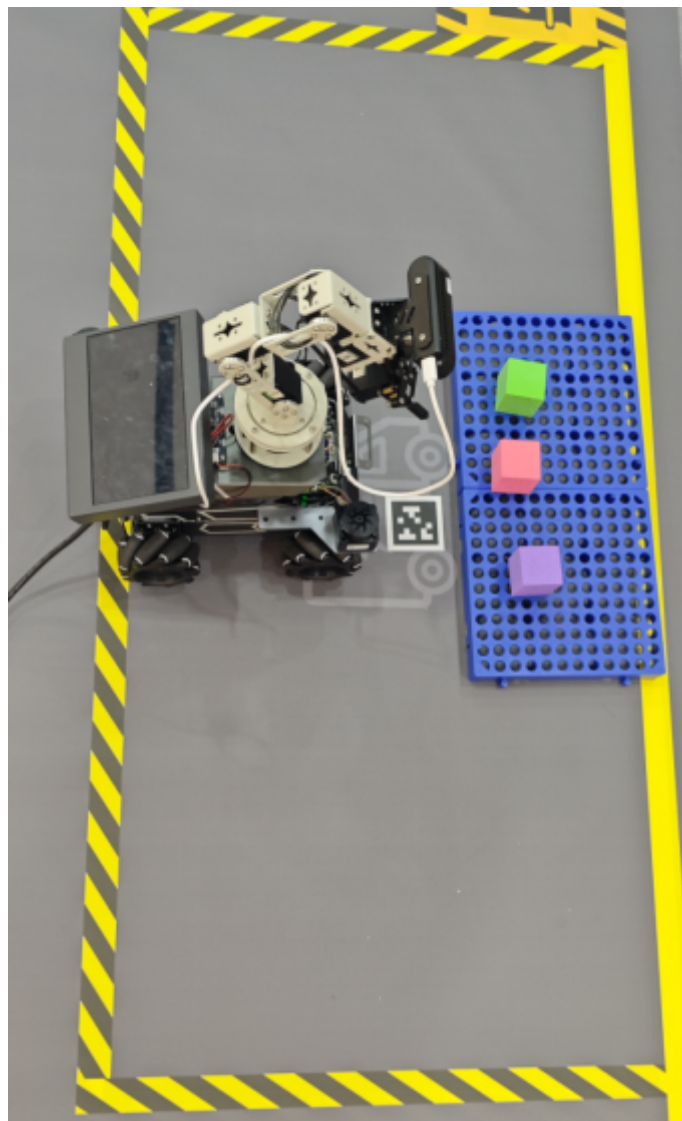


3.3 Cargo Inspection and Replenishment

- Check for missing cargo in the shipping area and replenish from the cargo warehouse.
- Cargo center

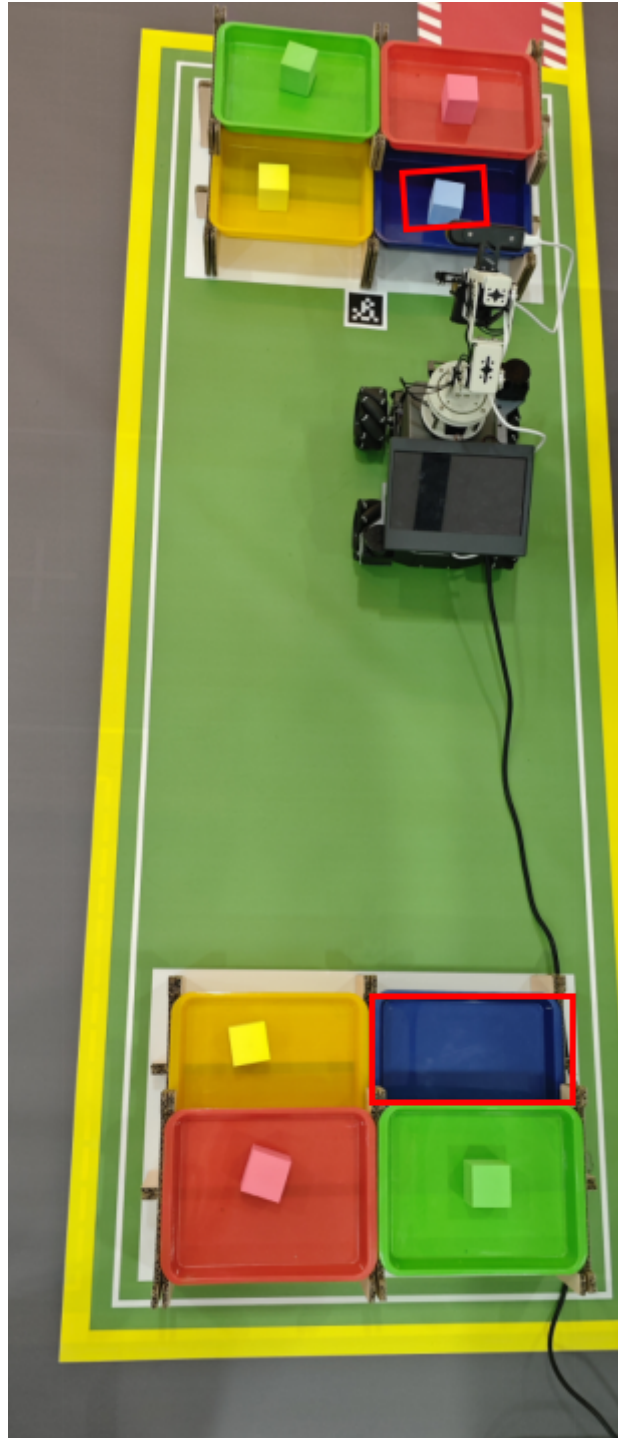


- Shipping area



3.4 Inter-Shelf Object Transportation

- Transport cargo from a designated shelf to a designated position on another shelf.



4. Comprehensive Case

- **You can arbitrarily combine the above cases**, or customize command cases according to actual needs. The following is a reference comprehensive case:
- Factory comprehensive case 1

[!IMPORTANT]

- The example long-flow task here takes a long time, approximately 25-40 minutes.

Start today's factory inspection. First, sort all garbage in the garbage area into the corresponding trash bins. Then go to the cargo sorting area and sort the cargo into the corresponding color areas on shelf 3. After that, go to the shipping area to check if any cargo is missing. If cargo is missing, replenish from the cargo warehouse. Finally, take a blue cargo from shelf 1 and deliver it to the corresponding color tray on shelf 2. After completing the task, return to parking lot 1.

- Choose any interaction method to send the command.